

ML: Linear Regression

WEEK – 10

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Outline

- Data Types in Machine Learning
- Linear Regression
- Linear Regression Example Google Colab

Data Types in Machine Learning

Before a model can learn — it needs to understand the KIND of data it's working with



NUMERICAL

Numbers you can calculate with

Age = 22, Price = 499.99

→ Used in ML for: Regression



CATEGORICAL

Labels — no math, just groups

Color = Red, Gender = Male

→ Used in ML for: One-hot encoding, grouping, Classification



ORDINAL

Ordered categories with ranking

Grade = A > B > C

→ Used in ML for: Label encoding, ordinal models

Numerical Data

TYPE 1

💡 **Numerical data** = any data that is a NUMBER. You can add it, subtract it, find its average — it makes mathematical sense.

DISCRETE (Counted)

Only WHOLE numbers — you **COUNT** these, you can't have half.

Cars in a parking lot	0, 1, 2, 3 …
Students in a class	25, 30, 40 …
Items ordered online	1, 2, 5, 10 …

⚠ *Can't have 2.5 cars or 30.7 students!*

CONTINUOUS (Measured)

Can be ANY number — including decimals. You **MEASURE** these.

Temperature today	36.6 °C
Price of a laptop	Rs. 85,499.50
Height of a student	5.87 ft

✓ *36.6, 36.61, 36.619 — infinite precision possible!*

Categorical & Ordinal Data

CATEGORICAL Data

Groups or labels — these are **NAMES**, not numbers. You cannot say one is 'greater than' another.

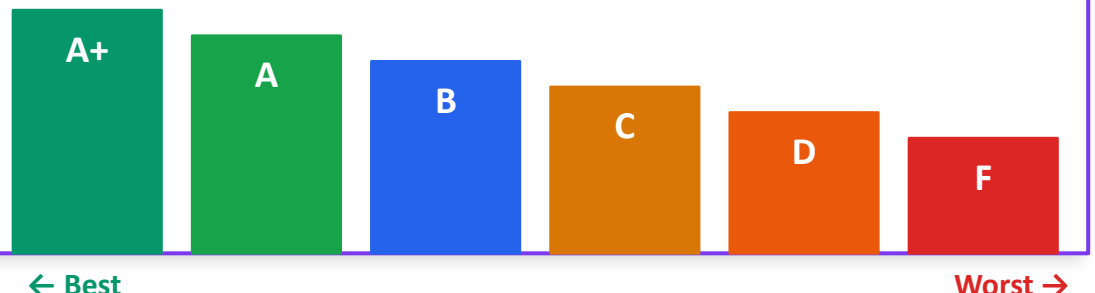
- Red / Blue / Green
- Male / Female
- Yes / No
- Cat / Dog / Bird
- City name
- Food type

Key insight: Is Red > Blue? No! They are just DIFFERENT — no ranking. ML uses one-hot encoding for these.

ORDINAL Data

Like categorical BUT with a clear **ORDER** / ranking. One value IS better or worse than another.

- Pain level: None < Mild < Moderate < Severe
- Movie rating: 1★ < 2★ < 3★ < 4★ < 5★



Summary — At a Glance

Data Type	Are they numbers?	Can they be ranked?	ML Technique
Discrete (Numerical)	✅ Yes (whole numbers)	✅ Yes (1 < 2 < 3)	**Use as-is or normalize**
Continuous (Numerical)	✅ Yes (any decimals)	✅ Yes (can compare)	**Normalize / standardize**
Categorical	❌ No	❌ No (just labels)	**One-Hot Encoding**
Ordinal	❌ No (but has order)	✅ Yes (A > B > C)	**Label / Ordinal Encoding**

🎯 **Why does this matter? ML algorithms treat each data type differently — wrong encoding = poor model performance!**

Regression

- Regression is used when you try to find the relationship between variables.
- In Machine Learning, and in statistical modeling, that relationship is used to **predict** the outcome of future events.

Regression

- **Dependent variable**: Whose values depend on the other variables
- **Independent variable**: Who forms the basis of estimation

Independent variables	Dependent variables
Income	Expenditure
Height of the parents	Heights of the children
My study hours	Marks in exams

Linear Regression

Understanding the equation $y = b_0 + b_1X$
step by step — with real examples



1. What is it?
2. The equation
3. What each part means
4. Worked example

What Is Regression? — The Big Idea

🎯 **Goal:** Given some input value X, predict the output value Y using a straight line.

Real-Life Analogy

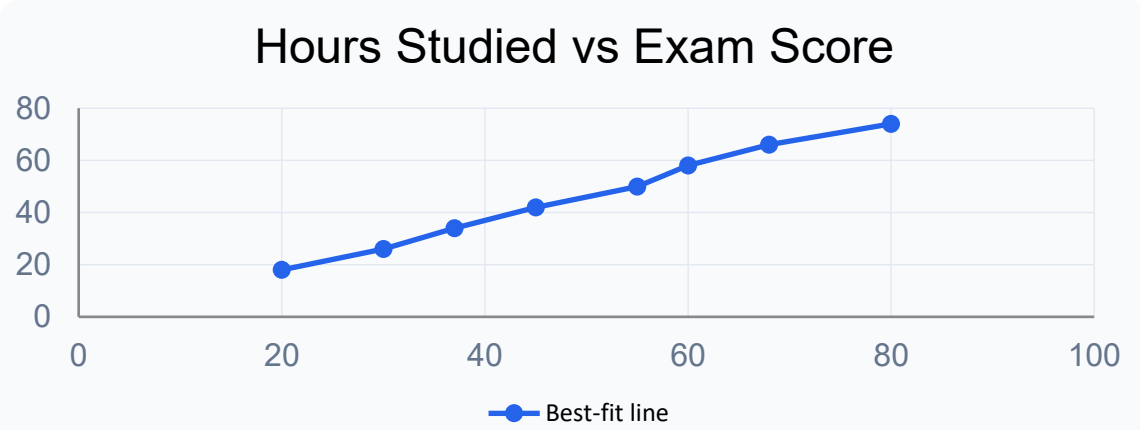
Imagine you want to **predict a student's exam score** based on **how many hours they studied.**

More hours studied → **Higher score.**
Regression finds EXACTLY how much higher.

X (Hours Studied) = INPUT known data
Y (Exam Score) = OUTPUT what we predict

What Regression Does

1. Takes existing data points
2. Draws the BEST-FIT straight line



The Equation

$$y = b_0 + b_1 X$$

y
Predicted output (what we want)

b_0
Intercept — starting point

b_1
Slope — rate of change

X
Input — what we already know

Think of it like this:

b_0 — Starting Point

A student who studied 0 hours still scored 30 marks (from prior knowledge).
$b_0 = 30$

b_1 — Rate of Change

Every 1 extra hour of study adds 5 marks to the score.
$b_1 = 5$

X — Your Input

You tell the model: 'This student studied 4 hours'.
$X = 4$

Understanding b_0 — The Intercept (Starting Point)

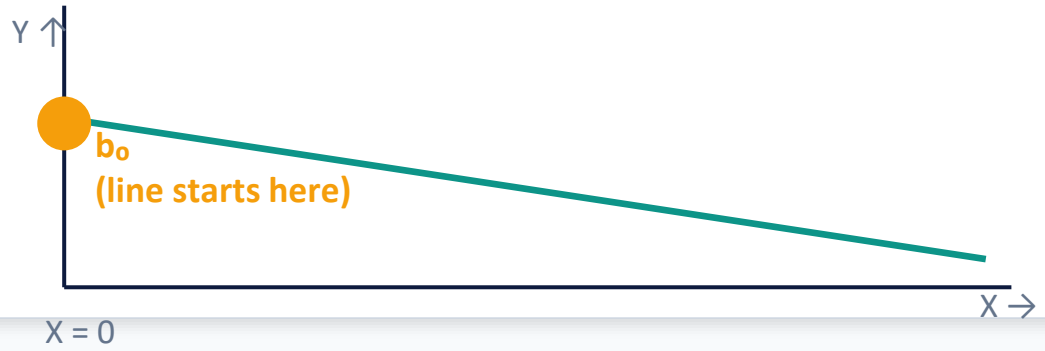
$y = b_0 + b_1X$ ← *We are focusing on this part today*

What is b_0?

b_0 is the value of Y when X equals ZERO.

It is where the regression line crosses the Y-axis — the starting value.

On the graph:



Real Examples of b_0

$b_0 = 30$

Student scores 30 marks with 0 hours studied

Base score from prior knowledge

$b_0 = 20$

House costs Rs. 20 lakh even with 0 sq.ft construction

Land value, basic structure cost

$b_0 = 50$

Taxi charges Rs. 50 before moving even 1 km

Base / flag-fall fare

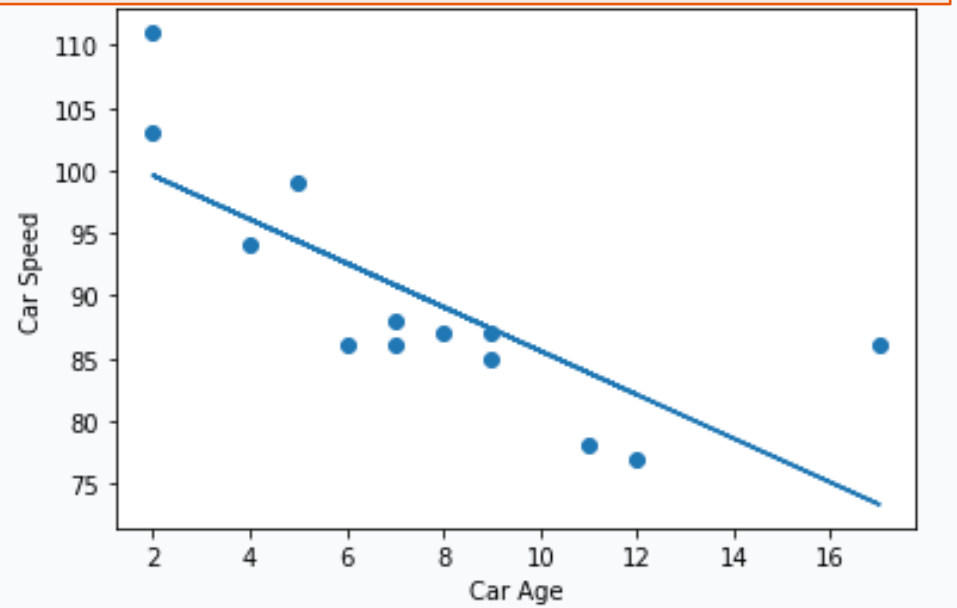
Understanding b_1 — The Slope (Rate of Change)

$y = b_0 + \mathbf{b_1X}$ ← *We are focusing on this part now*

What is b_1?

b_1 tells you: for every 1 unit increase in X, how much does Y change?

It is the STEEPNESS of the line.
Bigger b_1 = steeper line = stronger effect.



b_1 in Action — Every unit of X moves Y by b_1

Hours (X)	Score (Y)	Change	What this means
0 hours	30	—	b_0 = 30 — starting point, no studying
1 hour	35	+5	Each hour × b_1(5) = 5 more marks
2 hours	40	+5	Same — b_1 is constant each hour
5 hours	55	+5 per hour	5 extra hours × 5 = 25 extra marks
10 hours	80	+5 per hour	10 hours × 5 = 50 extra on top of b_0

The Variables — X and Y, and Their Many Names

X — The INPUT

X is the data you ALREADY HAVE before making a prediction.

Independent Variable

It doesn't depend on Y — it stands alone

Predictor

You USE it to make the prediction

Feature / Feature Vector

(ML term) One column of your data table

Example: Hours studied, house size, temperature

Y — The OUTPUT

Y is what you are TRYING TO PREDICT. You don't know it yet.

Dependent Variable

Its value DEPENDS on what X is

Response Variable

It RESPONDS to changes in X

Target

(ML term) What the model is trained to hit

Example: Exam score, house price, tomorrow's temperature

Summary — Your Linear Regression Cheat Sheet

Target / Output

y

What you are predicting.
Dependent variable.

Feature / Input

X

What you already know.
Independent variable.

Intercept

b_0

Value of Y when X = 0.
The starting point.

Slope / Coefficient

b_1

Change in Y per 1 unit increase in X.

The equation: $y = b_0 + b_1X$ ← *The best-fit straight line through your data*

Regression

- A statistical technique that **relates a dependent** variable to **one or more independent** variables.
- Most common goal in statistics is to answer the question
  - Is the variable **X associated** with a variable **Y**, and if so,
  - what is the **relationship** and can we **use it to predict Y**?

- This process of training a model on data where the outcome is known, for subsequent application to data where the outcome is not known, is termed **supervised learning**.

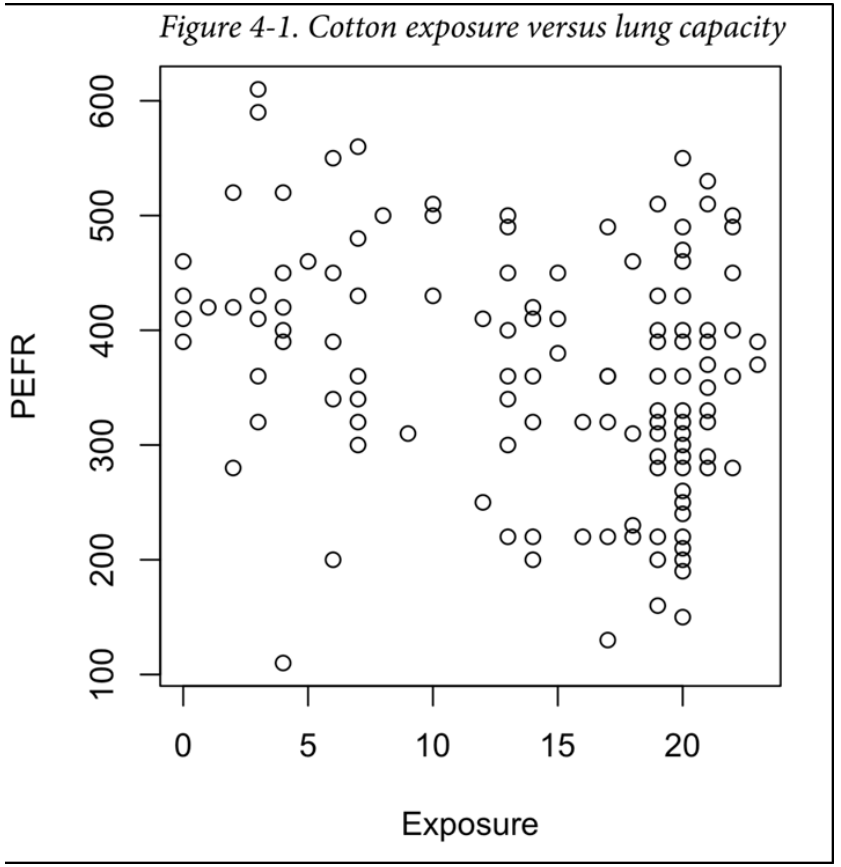
- Typically, a regression analysis is done for one of two purposes:
  - To **predict the value of the dependent** variable for individuals for whom some information concerning the explanatory variables is available, or
  - To **estimate the effect of some explanatory variable** on the dependent variable

Simple Linear Regression

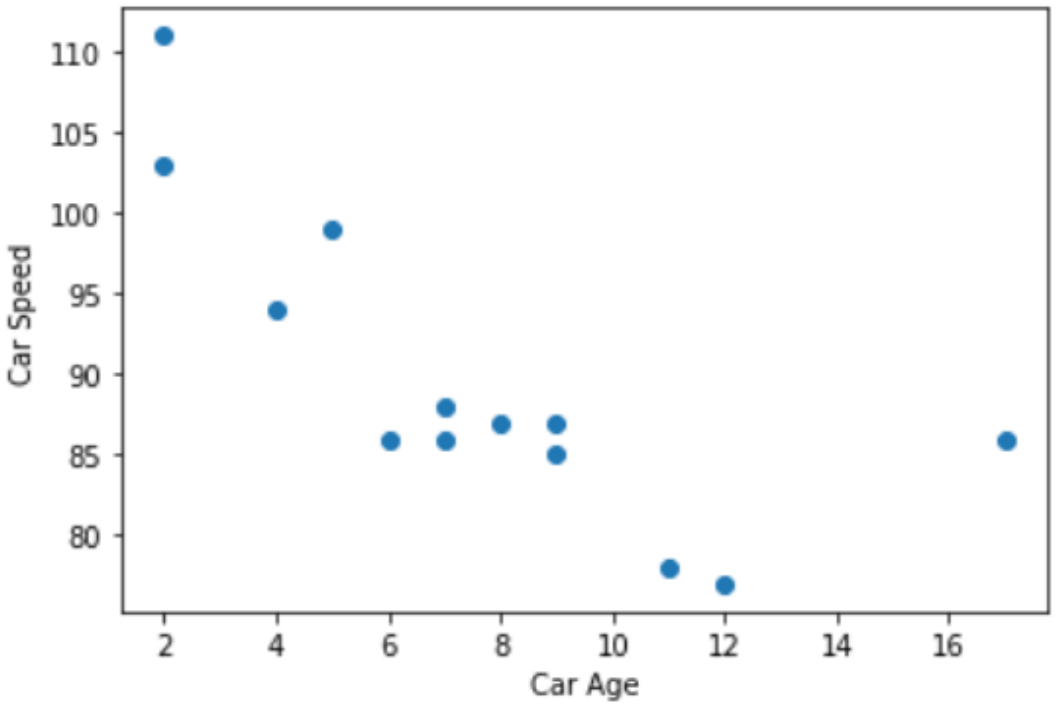
- Provides a model of the **relationship** between the **magnitude of one variable** and **that of a second**—for example, as X increases, Y also increases. Or as X increases, Y decreases.
- **Correlation** is another way to measure how two variables are related.
- The difference between regression and correlation is:
  - Correlation measures the **strength of an association** between two variables.
  - Regression **quantifies the nature of the relationship**.
- Simple linear regression estimates how much Y will change when X changes by a certain amount. With the correlation coefficient, the variables X and Y are inter-changeable.

Simple Linear Regression(The Regression Equation)

- Consider the scatterplot in **Figure 4-1** displaying the number of years a worker was exposed to cotton dust (**Exposure**) versus a measure of lung capacity (**PEFR** or "peak expiratory flow rate").
- How is **PEFR related to Exposure**?
- hard to tell based just on the picture.
- Simple linear regression tries to find the "**best" line** to predict the response PEFR as a function of the predictor variable Exposure:
  - **PEFR = b0 + b1 Exposure**



Scatter plot




```python
# In the example below,
# the x-axis represents car age,
# and the y-axis represents car speed.
# Here is the age and speed of 13 cars were passing a tollbooth.
# Let us see if the data we collected could be used in a linear regression:

import matplotlib.pyplot as plt

x = [5,7,8,7,2,17,2,9,4,11,12,9,6] # car age
y = [99,86,87,88,111,86,103,87,94,78,77,85,86] # car speed

plt.scatter(x, y)
plt.xlabel("Car Age")
plt.ylabel("Car Speed")
plt.show()
```

Line of Linear Regression

```python
import matplotlib.pyplot as plt
from scipy import stats

x = [5,7,8,7,2,17,2,9,4,11,12,9,6]
y = [99,86,87,88,111,86,103,87,94,78,77,85,86]

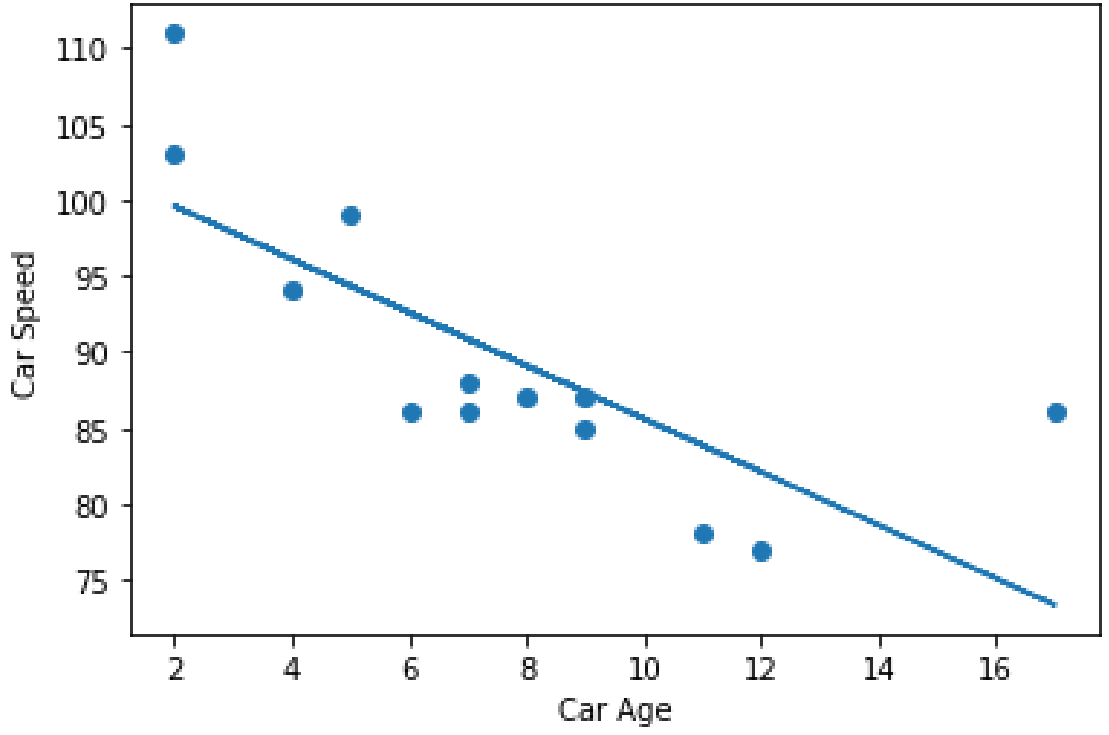
slope, intercept, r, p, std_err = stats.linregress(x, y)

# slope: The steepness of the line.
# intercept: The point where the line crosses the y-axis.
# r: Correlation coefficient (how well x predicts y).
# p: P-value (statistical significance).
# std_err: Standard error of the slope estimate.


def myfunc(x):
  return slope * x + intercept

mymodel = list(map(myfunc, x))

plt.scatter(x, y)
plt.plot(x, mymodel)
plt.show()
```



- Defines a function to compute predicted values (**y = mx + b**).
- **map(myfunc, x)**: Applies the function to each **x** value.

Regression Example (Paper pencil)

Formula for linear regression equation is given by:

$$y = a + bx$$

a and *b* are given by the following formulas:

$$a\,(intercept) = \frac{\sum y \sum x^2 - \sum x \sum xy}{(\sum x^2) - (\sum x)^2}$$

$$b\,(slope) = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

Where,

x and *y* are two variables on the regression line.

b = Slope of the line.

a = *y*-intercept of the line.

x = Values of the first data set.

y = Values of the second data set.

Regression Example (Paper pencil)

Question: Find linear regression equation for the following two sets of data:

x	2	4	6	8
y	3	7	5	10

Regression Example (Paper pencil)

x	y	x^2	xy
2	3	4	6
4	7	16	28
6	5	36	30
8	10	64	80
$\sum x$ = 20	$\sum y$ = 25	$\sum x^2$ = 120	$\sum xy$ = 144

$$b = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}$$

$$b = \frac{4\times 144 - 20\times 25}{4\times 120 - 400}$$

b = 0.95

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

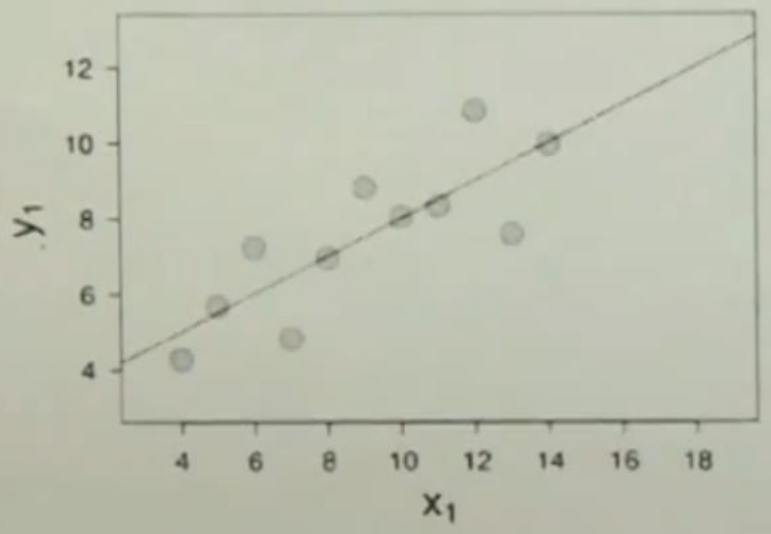
$$a = \frac{25\times 120 - 20\times 144}{4(120) - 400}$$

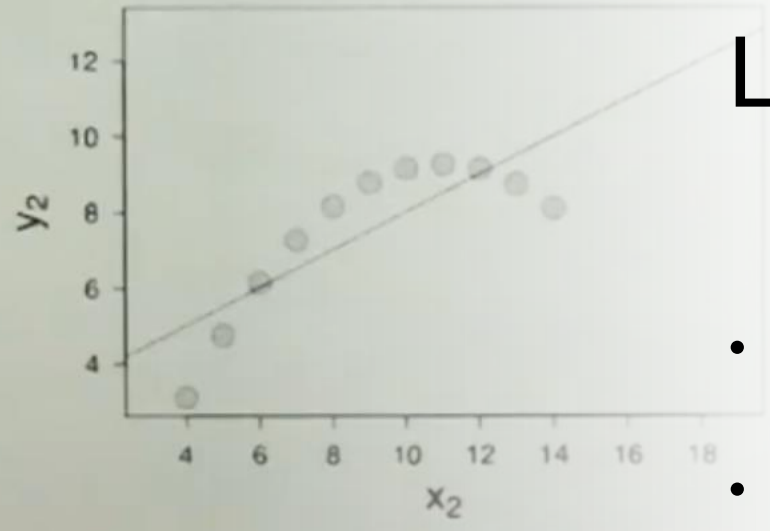
a = 1.5

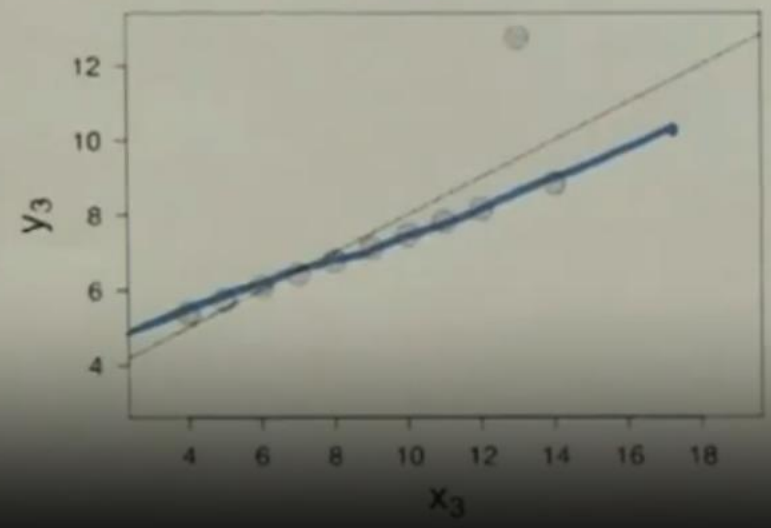
Linear regression is given by:

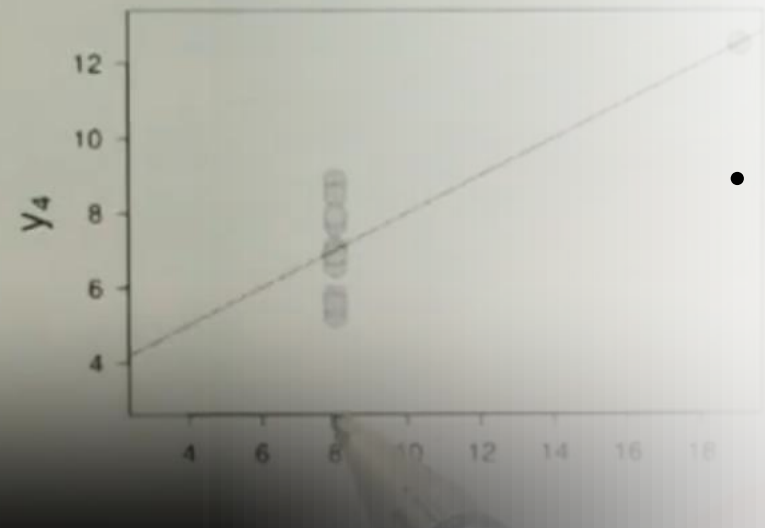
y = a + bx

y = 1.5 + 0.95 x









Linear Regression

- Linear regression is suitable when data is linear.
- Performs poor when there are:
  - Outliers in data
  - Non-linear data
  - y-axis may go infinity
- The solution is logistic regression